

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech

Branch : Chemical Engg.

Semester: III

Subject Code & Name: (BTCHC302) FLUID FLOW OPERATIONS

Max Marks: 60

Date:07/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question) **12**

1 Manometers are used for the measurement of _____

L2/CO1

1

a. atmospheric pressure	b. gauge pressure	c. difference in pressures	d. absolute pressure
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2 The unit of absolute viscosity is,

L2/CO1

1

a. centipoise	b. Pascal- sec	c. Kg/m sec	d. all
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3 The Continuity equation is,

L2/CO2

1

a. momentum balance equation.	b. mass balance equation.	c. both mass and momentum balance equations	d. energy balance equation.
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4 Friction factor is ratio of

L2/CO2

1

a. wall shear stress to viscosity	b. wall shear stress to product of density and viscosity	c. velocity head to density	d. wall shear stress to product of density and velocity head
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5 The friction loss coefficient for sudden expansion of cross section is,

L2/CO2

1

a. $(S_a/S_b - 1)^2$	b. $(S_a - S_b)^2$	c. $(1 - S_a/S_b)^2$	d. $(S_a/S_b - S_b)^2$
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1

- 6 The maximum flow the pump can deliver under any operating conditions is called

L2/C4

a. maximum head flow rate	b. minimum head flow rate	c. unit head flow rate	d. zero head flow rate
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1

- 7 In fluidization, at incipient of fluidization the porosity is

L2/CO3

a. constant	b. maximum	c. minimum	d. infinite
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1

- 8 The increased kinetic energy is converted in to pressure energy in _____ of centrifugal pump.

L2/CO4

a. volute	b. vanes	c. discharge pipe	d. eye of impeller
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1

- 9 To avoid cavitation in centrifugal pumps, the pressure at the pump inlet must

L2/CO4

a. be less than the vapor pressure of liquid by certain value	b. exceed the vapor pressure of liquid by certain value	c. be equal to the vapor pressure of liquid	d. be less than the atmospheric pressure
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1

- 10 A multistage centrifugal pumps has more than two _____

L2/CO4

a. Pumps	b. shafts	c. pistons	d. Impellers
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1

- 11 The pressure recovery in a well-designed venturimeter is about _____ of total pressure drop.

L2/CO5

a. 70%	b. 80%	c. 90%	d. 100%
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1

- 12 The vena-contracta is _____ cross section of jet after _____

L2/CO5

a. maximum ; sudden expansion	b. constant ; sudden expansion	c. maximum ; sudden contraction	d. minimum ; sudden contraction
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Q. 2	Solve the following.		12
A)	State the assumptions made while deriving the Bernoulli's equation. Write the momentum balance equation and calculate each term with reference to Bernoulli's theorem.	L2/CO1	6
B) i)	For following situations of steady flow determine whether flow is Laminar or Turbulent.	L3/CO1	6
	a) Water flows at average velocity 1.5 m/s through a 100 mm diameter tube. Viscosity = 1 cp. Density = 1000 kg/m ³		
	b) A polymer melt having density 900 Kg/m ³ and viscosity 0.25 Pa-Sec flows through 50 mm tube at velocity 0.50 m/s.		
ii)	Estimate the transition length for a 25 mm diameter tube through which water is flowing having viscosity 1 cp.		
	a) Velocity is 0.081 m/sec		
	b) Velocity is 1.1 m/sec		
Q.3	Solve the following.		12
A)	Derive the Hagen Poiseuille Equation and prove that $f = 16 / (N_{Re})$ for laminar flow through circular pipe..	L2/CO2	6
B)	Water is pumped to a height of 20 m from the water level in reservoir, through a pipe of 10 cm I.D. at an average velocity of 2 m/s. If the pipeline along with the fittings is equivalent to 250 m long and the overall efficiency is 70%, The friction factor is given by, $f = 0.0791 / ((Re)^{0.25})$	L3/CO2	6
	a) Calculate friction loss in piping system.		
	b) What is the power required for pumping?		
Q. 4	Solve Any Two of the following.		12
A)	What is fluidization? Explain superficial velocity and minimum fluidization velocity. Explain applications of fluidization	L2/CO4	6
B)	A bed of catalyst particles (Sp Gr 2.0) of uniform size 1500 Micro meters is in 0.5 m diameter and 0.5 m high bed. In the packed state the porosity may be taken as, 0.4. Ergon's equation for the system is as given below. $\frac{\Delta P}{L} = 375 \times 10^3 V_{om} + 10.94 \times 10^6 V_{om}^2$ If water is to be used for fluidization calculate minimum fluidization velocity. During fluidization at bed height of 1 m what will be the porosity of the bed.	L2/CO4	6

- C) What is drag? Define drag coefficient. Show and explain nature graph between drag coefficient and Reynolds number for typical shapes. L2/CO4 6

Q.5 Solve Any Two of the following. 12

- A) Classify different gas moving machineries. In what pressure rating they are used? Discuss equations to calculate power required for compressor. L2/CO5 6
- B) Differentiate between positive displacement and centrifugal pumps. Discuss the characteristic curves to study the performance of pump? L2/CO5 6
- C) Benzene is pumped at a rate $9 \text{ m}^3/\text{hr}$ from reservoir at atmospheric pressure to an overhead tank at 15 m height. The pump suction is 2.5 m from level in reservoir. If friction in suction line is 4.5 J/kg , and in discharge piping is 35.5 J/kg , Gauge pressure at the end of discharge pipe is 222.91 KPa, and entire pipe is 5 cm I.D., L3/CO5 6

$$\rho_{(\text{benzene})} = 860 \text{ Kg/m}^3$$

Vapour pressure of benzene = 15 Kpa,

pump efficiency is 60%

- a) Calculate Power required for pumping &
b) N.P.S.H. available of the pump.

Q. 6 Solve Any Two of the following. 12

- A) Differentiate between variable head meters and variable area meters. With neat sketch explain Rotameter. What are the different forces acting on the float of Rotameter? 6
- B) A pipe with 75 mm I.D. is carrying a solution in which a venturimeter is fitted having throat diameter of 36mm. The differential manometer filled with mercury connected across meter reads a level difference of 215 mm. Calculate volumetric flow rate of acid through pipe. 6

$$C_v = 0.98, \rho_{\text{Acid}} = 1200 \text{ kg/m}^3, \rho_{\text{Hg}} = 13600 \text{ kg/m}^3$$

- C) What are different types of impellers used in agitation system? Discuss effect of axial flow, radial flow and tangential flow developed by impellers. 6

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